

PNG Census 2025

Hybrid Mobile App — Summary & Architecture

Flutter · Isar · Android/iOS · Offline-first · Bilingual EN/TPI

Overview

A fully offline-capable hybrid census data collection application built for Papua New Guinea field enumerators. The app runs on Android and iOS, stores all data locally using an embedded Isar database, supports bilingual (English / Tok Pisin) question display, and synchronises records to a central server whenever connectivity is available. A supervisor dashboard provides real-time progress visibility and conflict resolution.

Build Phases

Phase	Deliverable
1 — Environment & Scaffolding	Project setup, dependencies, AndroidManifest permissions, asset structure
2 — Data Layer	Isar DB schema (HouseholdRecord, VectorClock embedded type), build_runner code generation
3 — Form Engine	ExpressionEvaluator (skip logic, 8 operators), ValidationResult, FormEngine ChangeNotifier, repeat-group a
4 — Enumeration UI	Tabbed section navigation, progress bar, bilingual toggle, GPS capture (Haversine geofence), camera/galler
5 — Sync & Translation	Vector-clock conflict resolution, HTTP sync engine, Claude Haiku translation fallback (3-tier: glossary → cac
6 — Supervisor Dashboard	Progress stats by status, conflict queue, conflict detail screen with approve / return-to-draft actions

Architecture

The app is structured in four layers. Each layer depends only on layers below it; the UI never accesses the database directly.

UI Layer

- EnumerationScreen — tabbed form with TabController / TabBarView, ListenableBuilder for efficient rebuilds
- SupervisorScreen — DefaultTabController, two tabs with AutomaticKeepAliveClientMixin
- ConflictDetailScreen — orange-themed read-only record view with approve / return actions
- QuestionWidget — routes by QuestionType; custom _RadioTile, _OptionSheet (search for >7 options), GpsCaptureWidget, AudioPlayButton
- RepeatGroupWidget — expandable household roster cards with per-instance skip logic

Form Engine

- FormEngine extends ChangeNotifier — single source of truth for all answers
- ExpressionEvaluator — evaluates showWhen conditions (eq/neq/gt/lt/gte/lte/contains/empty/notEmpty)
- Answer pruning — _pruneHidden() clears answers for newly-hidden questions on every setAnswer()
- exportAnswers() — merges flat + repeat-group answers into one Map
- completionPercent / isSectionComplete() — drives progress bar and tab checkmarks
- toggleLocale() — switches between EN and TPI, triggers full UI rebuild

Data Layer

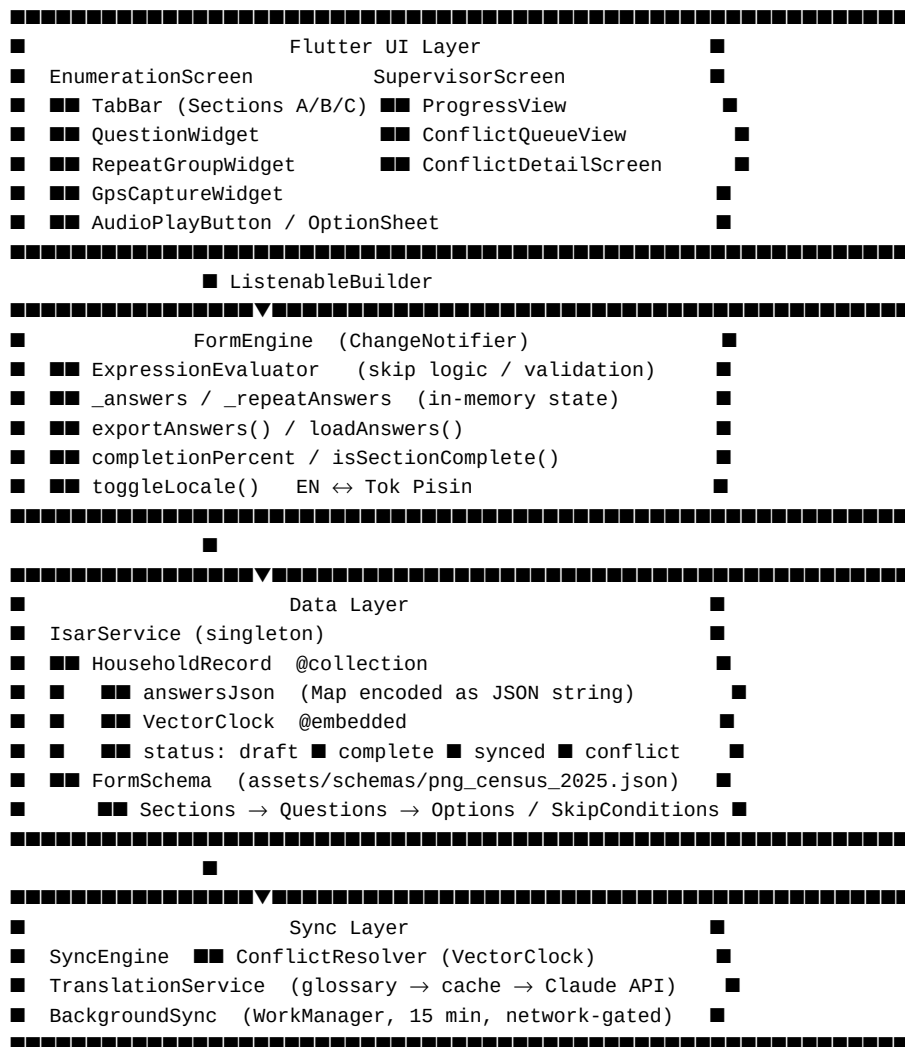
- IsarService (singleton) — wraps Isar 3.1, exposes typed CRUD helpers

- HouseholdRecord @collection — uuid (unique index), enumeratorId, status, answersJson (JSON blob), VectorClock
- VectorClock @embedded — two parallel lists (nodeIds, counters); increment(), merge(), happensBefore(), isConcurrentWith()
- RecordStatus enum — draft | complete | synced | conflict
- FormSchema — loaded from assets/schemas/png_census_2025.json; 3 sections, 43 questions total (incl. 22 roster sub-questions)

Sync Layer

- SyncEngine (singleton) — uploads complete/conflict records, fetches remote, resolves with ConflictResolver
- ConflictResolver — VectorClock comparison: if A happensBefore B → B wins; if concurrent → LWW merge, status = conflict
- TranslationService — 3-tier: built-in glossary (80 terms) → file-backed JSON cache → Claude Haiku API with prompt-caching beta header
- BackgroundSync — WorkManager periodic task (15 min, network-gated, exponential backoff); platform-guarded for Android/iOS only
- connectivity_plus — network state listener triggers immediate one-shot sync on reconnection

Layer Diagram



Census Form Schema

The form is defined in a single JSON asset (assets/schemas/png_census_2025.json) and loaded at runtime. Adding new questions or sections requires only a JSON update — no code change.

Section	Questions	Notable Features
A — Location & Identification	10	Province selector (all 22 provinces), District, LLG, Ward, Census Unit, Village, Household Number, HoH Name, GPS capture
B — Household Roster	1 repeat group (22 sub-questions per member)	Name, Relationship, Sex, Age, DoB, Marital Status (age≥12), Religion, Language, Citizenship, Education, Literacy, Employment, Occupation, Fertility — all with skip logic
C — Dwelling Characteristics	10	Type, Tenure, Sleeping Rooms, Floor/Wall/Roof materials, Water source, Toilet type, Cooking fuel, Lighting

Key Design Decisions

Offline-first — Isar embedded database stores all records locally. Sync is opportunistic — the app works fully without any network connection.

Vector Clocks — Distributed conflict detection without a central authority. Each node (enumerator device, supervisor) increments its own counter. Causality comparison determines which record wins; concurrent edits are flagged for supervisor review.

JSON Answer Blob — Answers are stored as a single JSON string in answersJson. This allows flexible schema evolution — adding or removing questions does not require a database migration.

Answer Pruning — _pruneHidden() runs on every setAnswer() call and clears answers for questions that become hidden due to skip logic. This prevents stale data from reaching the submitted record.

3-Tier Translation — The built-in glossary covers common census terminology instantly. The file-backed cache avoids repeat API calls. Claude Haiku with prompt-caching is the fallback — the large system prompt is cached server-side, keeping per-request cost low.

Platform Guards — WorkManager (background sync) is Android/iOS only. All calls are wrapped in Platform.isAndroid || Platform.isIOS so the app also builds and runs on Linux desktop for development and testing.

Dependencies

Package	Version	Purpose
isar + isar_flutter_libs	^3.1.0+1	Embedded NoSQL database (offline storage)
isar_generator + build_runner	^3.1.0+1 / ^2.4	Isar schema code generation
provider	^6.1.2	ChangeNotifier state management
connectivity_plus	^6.1.4	Network state detection
geolocator	^13.0.2	GPS position with LocationSettings API
image_picker	^1.1.2	Camera / gallery photo capture
audioplayers	^6.1.0	Audio question prompt playback
workmanager	^0.9.0	Background sync (Android/iOS)
http	^1.2.2	HTTP client for sync and Claude API
crypto	^3.0.6	SHA-256 cache key hashing for translations

path_provider	^2.1.5	Application documents directory
easy_localization	^3.0.7	i18n infrastructure